

Deep Learning (Assignment Project)

Handwritten Digit Classification using CNN & RNN

Project Dataset: MNIST Digit Classification

Problem Statement

Handwritten digit recognition is a fundamental problem in computer vision with applications in postal automation, bank cheque processing, and digitising handwritten documents. Traditional machine learning methods require manual feature extraction, which can limit performance and scalability.

The objective of this project is to develop a **Convolutional Neural Network (CNN)** and an **Recurrent Neural Network (RNN)** capable of automatically learning spatial features from handwritten images and accurately classifying digits from **0–9** using the **MNIST dataset**. We'll then compare the performance of both these neural networks.

The dataset contains grayscale images of handwritten digits, each of size **28 × 28 pixels**. The goal is to train a CNN model & an RNN model that can correctly classify each image into one of the ten digit classes.

The project involves designing an appropriate CNN & RNN architecture, training the model, evaluating its performance using metrics such as **accuracy and confusion matrix**, and analysing how different architectural choices affect classification performance.

Expected Outcome

A trained CNN & RNN model capable of recognising handwritten digits with high accuracy (typically >95%).